

E.3 / PAPER 1 / QUESTION 13

29 April 2025 / TZ2 (file label) / HL Paper 1A / original Q36 / 2225-6713

13. A fusion reaction of one nucleus of hydrogen-2 and one nucleus of hydrogen-3 converts a mass of 0.024 u into energy. A fission reaction of one nucleus of uranium-235 converts a mass of 0.48 u into energy.

What is $\frac{\text{energy released per unit mass of hydrogen}}{\text{energy released per unit mass of uranium}}$?

- A. 0.050
- B. 0.426
- C. 2.35
- D. 20.0